likely to have sucked a surplus of blood. On a subsequent night the luck may run the other way. So, it looks like a promising case for a bit of reciprocal altruism. Wilkinson found that those individuals who struck lucky on any one night did indeed sometimes donate blood, by regurgitation, to their less fortunate comrades. Out of 110 regurgitations that Wilkinson witnessed, 77 could easily be understood as cases of mothers feeding their children, and many other instances of blood-sharing involved other kinds of genetic relatives. There still remained, however, some examples of blood-sharing among unrelated bats, cases where the 'blood is thicker than water' explanation would not fit the facts. Significantly the individuals involved here tended to be frequent roostmates—they had every opportunity to interact with one another repeatedly, as is required for an Iterated Prisoner's Dilemma. But were the other requirements for a Prisoner's Dilemma met? The pay-off matrix in Figure D is what we should expect if they were.

		What you do	
		Cooperate	**Defect**
What I do	**Cooperate**	Fairly good **REWARD** I get blood on my unlucky nights, which saves me from starving. I have to give blood on my lucky nights, which doesn't cost me too much.	Very bad **SUCKER'S PAY-OFF** I pay the cost of saving your life on my good night. But on my bad night your don't feed me and I run a real risk of starving to death.
	Defect	Very good **TEMPTATION** You save my life on my poor night. But then I get the added benefit of not having to pay the slight cost of feeding you on my good night.	Fairly bad **PUNISHMENT** I don't have to pay the slight costs of feeding you on my good nights. But I run a real risk of starving on my poor nights.

Figure D. Vampire bat blood-donor scheme: pay-offs to me from various outcomes

Do vampire economics really conform to this table? Wilkinson looked at the rate at which starved vampires lose weight. From this he calculated the time it would take a sated bat to starve to death, the time it would take an empty bat to starve to death, and all intermediates. This enabled him to cash out blood in the currency of hours of prolonged life.